

Use Cases for Agentic Portal

Use Case 1: Multi-Chatbot Intelligent Routing and Knowledge Integration

- **Use Case ID:** UC-01
- **Use Case Name:** Multi-Chatbot Intelligent Routing and Knowledge Integration
- **Level:** User Goal
- **Primary Actor:** Enterprise Employee (or System User)
- **Stakeholders and Interests:**
 - User: There are multiple vertical domain Chatbots within the company (information silos). Users have to manually decide which platform to ask, or ask the same question to multiple chatbots to get information from various aspects. As the number of Bots increases, the memory burden on users is extremely high, and information retrieval efficiency is low. Users hope to get answers quickly and accurately without needing to care about which bot(s) are behind the scenes.
- **Preconditions:**
 - Each bot provides an API interface for invocation and specifies user permission requirements. The user has successfully logged into the Agent Portal.
- **Main Success Scenario (Basic Flow):**
 1. The user enters a question in natural language into the unified input box (e.g., "I want to know about the data samples for the Pilum system and the specific execution process of the MPU department.").
 2. The system analyzes the question, extracts key intents and entities, and determines whether the question requires cross-domain collaboration.
 3. Based on the analysis results, the system matches routing rules and selects one or more most suitable specialized Bots (e.g., selecting both PuG ChatGPT and MPU Bot simultaneously).
 4. The system concurrently forwards the user's question (along with necessary contextual information) to the API interfaces of the relevant Bots.

5. The system receives the returned content from each Bot and uses an internal LLM to perform knowledge aggregation and conflict detection based on the original intent, generating a logically unified final answer to display to the user.
 6. The user receives the answer, and the use case ends.
- **Exception Flows:**
 - **3a: Unable to Match a Suitable Specialized Bot**
 1. After identifying the intent, the system determines it does not fall under any configured professional domain.
 2. The system invokes a general-purpose LLM (e.g., the company's general ChatGPT) to answer the question and prompts the user that the question does not pertain to a specific business domain.
 - **Postconditions:**
 - The system records the user's query and the corresponding response results from the bots (for logging/analysis). The user receives a satisfactory answer.
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Use Case 2: Cross-Project AI Information Integration and Comprehensive Analysis

- **Use Case ID:** UC-02
- **Use Case Name:** Cross-Project AI Information Integration and Comprehensive Analysis
- **Primary Actor:** End User (e.g., Data Analyst, Business Personnel)
- **Brief Description:**

Users initiate complex queries through the unified Agent Portal. The system utilizes the LLM's Tool Use capability to invoke APIs of multiple independent AI projects via an MCP bridge, performing cross-comparison, mutual enhancement, or information fusion on the returned results to present a comprehensive conclusion.
- **Preconditions:**
 1. The user has successfully logged into the Agent Portal.

2. The MCP services of the relevant AI projects are online and have been registered.
3. The relevant AI projects provide API interfaces.

- **Basic Flow:**

1. The user enters a query involving multiple projects into the Portal's unified interaction box (e.g., "I want to know the price reduction strategy for part A, and also check if the company producing part A has any financial risks, so I can negotiate with Company A.").
2. The Portal's built-in LLM parses the user's intent, identifying multiple tools required to complete the task (e.g., RPP Pro, Financial Risk Analysis, Nego Pro).
3. The LLM sends tool invocation instructions via the MCP protocol.
4. The MCP Bridge invokes the API interfaces of each independent AI project based on the instructions.
5. Each AI project returns structured processing results to the MCP Bridge.
6. The LLM receives the results from all parties and performs data processing (e.g., comparing differences, supplementing information, logical reasoning).
7. The system displays the final fused response on the interface, indicating the sources of information.

- **Alternative/Exception Flows:**

- **1a. Tool Matching Failed:** The LLM cannot find an AI project that meets the requirements and prompts the user to narrow down the query scope.
- **4a. Interface Call Timeout:** A specific AI project times out; the system indicates that the project is unavailable and generates results based on the remaining projects.
- **6a. Information Conflict:** Significant contradictions exist between results from different projects; the system highlights the conflicting points for the user's reference.

- **Postconditions:**

The system successfully displays the comprehensive analysis results and logs the execution chain of this cross-project collaboration in the background.

Use Case 3: Dynamic AI Project Expansion and Integration

- **Use Case ID:** UC-03

- **Use Case Name:** Dynamic AI Project Expansion and Integration

- **Primary Actor:** System Administrator / Developer

- **Brief Description:**

When a new AI project needs to be integrated, there is no need to modify the core code of the Agent Portal. The developer only needs to provide a standard API and add the corresponding configuration entry in the MCP server. The Portal can then automatically detect the new project and allow users to invoke its functionality.

- **Preconditions:**

1. The new AI project's API interface is deployed and accessible.
2. The administrator has permission to modify the MCP server configuration.

- **Basic Flow:**

1. The developer defines the input and output specifications for the new AI project's API.
2. The administrator adds a new entry for the project in the MCP server configuration file, including the tool name, functional description, and invocation address.
3. The administrator performs a hot update or restart of the MCP service to publish the new tool definition.
4. The Agent Portal automatically synchronizes the latest tool list via the MCP protocol.
5. The Portal's built-in LLM automatically understands the application scenario by reading the semantic description of the newly added tool.
6. The administrator performs a functional test on the Portal interface to verify whether Tool Use can correctly trigger the new project's API.

- **Alternative/Exception Flows:**

- **2a. Inappropriate Semantic Description:** The LLM fails to invoke the tool correctly due to a vague tool description; the administrator needs to optimize the description text.
- **3a. Configuration Format Error:** The MCP entry does not conform to the JSON Schema specification; the system indicates a loading failure.
- **Postconditions:**

The new AI project is successfully integrated into the platform. Users can immediately invoke the project's functionality through natural language in the Portal.